

When Set Scores Fail to Select Evidence: A Controlled Study of Budgeted Multi-hop RAG

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Abstract. Selecting complementary evidence under a limited context budget requires a score that remains useful on the sets found by search. We examine this requirement with first-, second-, and fourth-order factorization models and a permutation-invariant set model, sharing dense candidates and support-annotation supervision. Strong performance on constructed set pairs does not translate into stronger evidence selection: the static fourth-order model loses complete annotated support relative to a size-matched dense baseline. A bounded correction trains the deployed score on actual search frontiers and final sets, with an equal-update static replay control. On the same 128-question HotpotQA/MuSiQue development panel, the primary fourth-order model’s annotated support completeness rises from 55/128 to 57/128, while answer F1 falls from 0.4181 to 0.3766, below maximal marginal relevance’s 0.4303. In both training seeds, every aligned model remains below this baseline in F1 point estimate. Exact granular-ball and k-means indexes preserve flat-search selections but do not accelerate the measured 128-candidate search. The results identify a concrete gap between a score’s ability to distinguish constructed sets and its ability to select reader-useful evidence. The intervention and index comparisons characterize this gap on exposed development data; they are not independent confirmation.

Keywords: Evidence selection · Multi-hop QA · Search alignment

1 Introduction

Multi-hop question answering often depends on a small combination of passages rather than the single passage most similar to the question. Under a limited reader budget, evidence selection must balance relevance, redundancy, and coverage. Set-wise selection makes this collective objective explicit [3], while beam retrieval preserves multiple partial evidence hypotheses [11]. Neither modeling interactions nor searching more combinations ensures that the resulting context will help a reader.

We investigate a particular failure mode: a learned evidence-set score can perform well on constructed training comparisons while assigning high scores to poorer sets encountered during deployment. This motivates a controlled study with frozen embeddings, shared candidates, a fixed reader, and four comparable set models. The first-, second-, and fourth-order models test the value of explicit

interactions; a DeepSets control tests whether any benefit requires their particular factorization.

We contribute a controlled account of this mismatch. First, a high-accuracy set scorer discards annotated support even though a size-matched dense set is available to its final decision. Second, search-aligned continuation changes support and answer scores in opposite directions on the same question panel, with different responses from low-order and general-set controls. Third, exact indexes preserve the resulting beam choices without speeding up this 128-candidate workload. The intervention follows established search-aware learning [7]; the contribution is the measured behavior under shared evidence and reader conditions, not a new factorization algebra or tree bound.

2 Budgeted Evidence Selection

Shared task. For question q , BGE-large-en-v1.5 [8] retrieves a common candidate set V_q of at most 128 whole paragraphs from a corpus assembled from exposed development data. A context contains $S \subseteq V_q$, $|S| \leq 6$, rendered in original BGE order. The actual tokenized reader input, including the question and instructions, must not exceed 1,024 tokens. This token constraint is distinct from the six-block constraint. No support paragraph is inserted into evaluation candidates.

Let G_q denote the known support annotations and $C(S)$ the annotations covered by the selected original passages. Supervision is

$$c(S) = |C(S) \cap G_q|/|G_q|, \quad f(S) = \mathbf{1}[G_q \subseteq C(S)]. \quad (1)$$

These labels measure annotated support, not factual truth or exact answer utility. Unannotated passages may still be useful. Training can use known support references, whereas deployment selection receives only the question, passages, embeddings, and token lengths.

Set scores and search. Query–passage features combine frozen normalized embeddings, their product and absolute difference, cosine similarity, and log length. A shared 128-dimensional network produces unary factors a_i and 32-dimensional interaction factors v_i . For order $H \in \{1, 2, 4\}$, the deployed score is

$$g_H(S) = b_q(|S|) + \frac{1}{6} \sum_{i \in S} a_i + \sum_{h=2}^H \frac{w_{q,h}^\top e_h(\{v_i : i \in S\})}{32 \binom{6}{h}}, \quad (2)$$

where e_h is the coordinatewise elementary symmetric polynomial. Its recurrence is the standard distinct-member ANOVA computation from higher-order factorization machines [1]. The DeepSets control replaces the interaction sum with an MLP of the summed passage features and query representation [10]. Each model has a full-support logit and an auxiliary coverage head.

Width-four beam search evaluates feasible extensions through depth six, retains visited sets, and includes a feasible Dense-K6 set in the final comparison.

It returns the highest-scoring visited set, with token length then stable IDs resolving final ties. Beam search is not a global optimum. Dense-budget and MMR-budget may retain more than six short blocks under the same token limit; Dense-K6 and MMR-K6 isolate cardinality in support analyses. MMR [2] uses the fixed score $0.7 s(q, i) - 0.3 \max_{j \in S} s(i, j)$.

Search-aligned continuation. Static supervision contains support sets, missing-support variants, same-size replacements, redundant additions, and random sets. We continue each model on a shared pool mined from actual FIT search: model final sets, partial frontiers, Dense-K6/MMR-K6, and limited replacements. Pools are merged across all four architectures, deduplicated, and capped at 64 sets per question. Reference support can augment static FIT examples but never the mined candidate universe.

For $u(S) = (f(S) + c(S))/2$ and $u(S^+) > u(S^-)$, the added preference loss is

$$\ell(S^+, S^-) = \text{softplus}(0.2[u(S^+) - u(S^-)] - [g(S^+) - g(S^-)]) . \quad (3)$$

Final pairs prioritize equal cardinality and similar length; frontier pairs prioritize the same parent and depth. Thus coverage differences between two incomplete sets supervise the score used by search directly. The objective is $L_{\text{base}} + 0.5L_{\text{final}} + 0.5L_{\text{frontier}}$, with query-normalized losses and equal static/mined base sampling. H4 static replay receives the same optimizer settings and number of updates, without mined examples or the added preferences; it does not match all forward-pass computation. Search-aware training and overestimation control have established precedents [7,5]; our intervention tests their relevance to this annotation-supervised selection problem, without borrowing their performance or theoretical guarantees.

3 Experimental Setting

We reuse previously exposed HotpotQA and MuSiQue materials [9,6]. Paragraph reconstruction yields 9,791 and 31,130 blocks. The 4,000 questions are divided into 3,166 FIT, 738 TUNE, and 96 historical-development questions (Table 1). Query groups remain separate, but FIT and TUNE share 57 HotpotQA and 4,753 MuSiQue documents. This is development-corpus retrieval, not full-Wikipedia or unseen-document evaluation. The 738-query selection analysis contains 196 HotpotQA and 542 MuSiQue questions. Top128 reaches all known support for 190 and 367, respectively. Nine MuSiQue questions have known support that cannot all fit the six-block/1,024-token constraint. All remain in the denominator.

The continuation mines 256 FIT queries per dataset and selects checkpoints using 64 non-QA TUNE queries per dataset. Hash-based sampling preserves existing question-type proportions and query groups. Up to two eight-epoch rounds use AdamW (learning rate 3×10^{-4} , weight decay 10^{-4} , batch eight queries, gradient norm capped at one). At epochs 2, 4, and 8, checkpoint selection uses dataset-equal actual support completeness, then coverage, original development

Table 1. Data roles (HotpotQA/MuSiQue question counts). MINE is within FIT; DEV_SELECT and QA are disjoint subsets of TUNE. All are exposed development material. Repeated seeds reuse these same questions.

Role	HotpotQA	MuSiQue	Use
FIT	748	2,418	Supervised training
TUNE	196	542	Static model development
Historical DEV	56	40	Earlier interface development
MINE	256	256	Search-state collection
DEV_SELECT	64	64	Continuation checkpoint choice
QA	64	64	Fixed reader evaluation

loss, and earlier epoch. QA never selects the checkpoint. The replay checkpoint follows the selected H4 update count. Both seeds (1729/2026) complete two rounds; the second seed was triggered by selection-panel improvement before new QA scores were observed. The intervention changes training examples, ranking loss, and checkpoint selection together; it is not a loss-only ablation.

The fixed QA panel contains 64 questions per dataset: 48 bridge and 16 comparison HotpotQA questions, and 32 two-hop and 32 longer-hop MuSiQue questions. The original FP16 Qwen2.5-3B-Instruct reader [4] uses greedy decoding, at most 128 output tokens, and a shared constrained single-answer-field protocol. Dataset-specific canonical EM/F1 score that field without answer-dependent cleaning. Identical full token/model/decoder identities reuse outputs; this is not a new generation. Failures remain in every mean. All panel questions are exposed development data. We report paired descriptive comparisons without new confirmation tests or significance claims. Three original training seeds are not three independent datasets.

4 Results

Good proxy fit can coexist with support loss. Table 2 contrasts constructed-pair accuracy with actual selection. H4 correctly orders most sampled complete/incomplete pairs, yet selects less complete support than Dense-K6 on both datasets. Relative to Dense-K6, it loses 27 and 42 complete-support queries while recovering only 5 and 24. Its mean sigmoid outputs on selected sets are 0.992 and 0.967, against actual completeness 0.709 and 0.223. These outputs are not calibrated probabilities. The discrepancy motivates training on states produced by search rather than improving only an auxiliary prediction metric.

Checkpoint-selected support improves. On DEV_SELECT, primary-seed H4 completeness increases from 0.4531 to 0.5156. H1, H2, and DeepSets increase by 0.0781, 0.0625, and 0.0625. These are checkpoint-selected development effects. All eight aligned checkpoints across the two seeds come from round one; round two did not improve the selection criterion. This panel is distinct from QA.

Table 2. Original static models (seed 1729): constructed-pair accuracy and actual complete support on all 738 TUNE questions. H/M denote HotpotQA/MuSiQue.

Method	Pair H	Pair M	Full H	Full M
Dense-K6	–	–	.8214	.2565
H1	.9475	.8978	.6531	.2454
H2	.9638	.9422	.6429	.2325
DeepSets	.9620	.9482	.6990	.2306
H4	.9645	.9464	.7092	.2232

Table 3. Same QA panel (64 questions per dataset), seed 1729. Full is annotated support completeness, with counts out of 128. F1 is dataset-equal canonical answer F1; Δ F1 is relative to MMR. Dense/MMR share the token ceiling but may use more than six blocks.

Method	Full (count)	F1	Δ F1
Dense	0.5000 (64)	0.4055	-0.0248
MMR	0.4766 (61)	0.4303	+0.0000
Static H4	0.4297 (55)	0.4181	-0.0122
H4 replay	0.4297 (55)	0.4046	-0.0257
H1 aligned	0.4766 (61)	0.4065	-0.0238
H2 aligned	0.4375 (56)	0.4057	-0.0245
DeepSets aligned	0.4688 (60)	0.4090	-0.0213
H4 aligned	0.4453 (57)	0.3766	-0.0537

Answer quality remains a separate endpoint. Table 3 retains Dense, MMR, the original H4 checkpoint, replay, and every aligned architecture on the same 128 questions. In the earlier three-seed experiment, H4’s mean dataset-equal F1 was 0.4171 (seed standard deviation 0.0139), compared with Dense’s 0.4055 and MMR’s 0.4303. The H4 range is 0.4027–0.4304; seed variation is neither a query confidence interval nor an ensemble result. Table 3 puts annotated support and answers on the same QA panel. For seed 1729, H4 completeness changes from 55/128 (0.4297) to 57/128 (0.4453), while F1 falls from 0.4181 to 0.3766. The paired support counts are four gains and two losses; answer F1 improves on 10 questions, falls on 14, and is unchanged on 104. These are marginal changes, not evidence that adding support causes an answer loss. For seed 2026, H4 completeness changes from 53/128 to 57/128 and F1 from 0.4304 to 0.3831. Aligned H1, H2, and DeepSets score 0.4126, 0.4173, and 0.3521 in that seed. Every method retains all 128 exposed questions. Neither seed establishes an F1 advantage over MMR. The token ceiling is shared, but actual input lengths and block counts differ: Dense/MMR average 1,009.9/1,007.3 tokens, versus 916.9/842.2 for static/aligned H4. This is a pipeline answer comparison; the size-matched support diagnostic in Table 2 answers a different question.

Exact selection does not imply acceleration. The original exact granular-ball and k-means indexes matched Flat on 1,476 natural comparisons across 738 questions. In the subsequent fixed 32+32-query timing comparison, Flat took 0.335/0.546 seconds per question, granular balls 0.426/0.653, and k-means 0.423/0.644. Granular balls reduced candidate inner products by about 66%/44%, but increased token-feasibility work and retained index overhead. These measurements include per-query construction and network work; some ran alongside GPU reading, so they are not isolated hardware benchmarks. We preserve this no-speedup finding and do not repeat index tuning in the present intervention. The bounded continuation uses 1.25 GPU process-wall hours and 986 actual reader calls, including eight uncached checks. Identical generation identities are reused; the 1,536 logical rows are not independent samples or fresh calls.

5 Discussion and Conclusion

Evidence selection has multiple objectives that cannot be substituted for one another. Annotation supervision makes set prediction inexpensive, but does not certify answer utility. Search can expose a mismatch that sampled set comparisons miss. Exact indexing can preserve a poor choice perfectly, and fewer inner products can still cost more time. The bounded continuation therefore evaluates actual selected support and a fixed reader rather than declaring success from loss reduction alone. H2 improves slightly over its own static checkpoint in both seeds; H1 improves in only the primary seed. All aligned models remain below MMR in F1 point estimate, and H4 declines in both seeds. These controls support a conditional analysis of training alignment, not an exclusive advantage of higher interaction order.

These findings are bounded by exposed development data, document overlap, incomplete support annotation, one reader, and 128-candidate retrieval. They establish neither open-domain gains nor an exclusive granular-ball benefit. Independent confirmation requires a prospective data boundary.

Reproducibility and disclosure. Anonymous supplementary materials provide per-query numeric outcomes, portable table reconstruction, and data/model preparation specifications. They do not redistribute benchmark text, answers, or model weights, and do not constitute a complete model-rerun distribution. AI assistance covered implementation, analysis, literature checking, drafting, and revision, beyond language polishing.

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